

CloudSync AI Support System

Week 1 Assignment Submission

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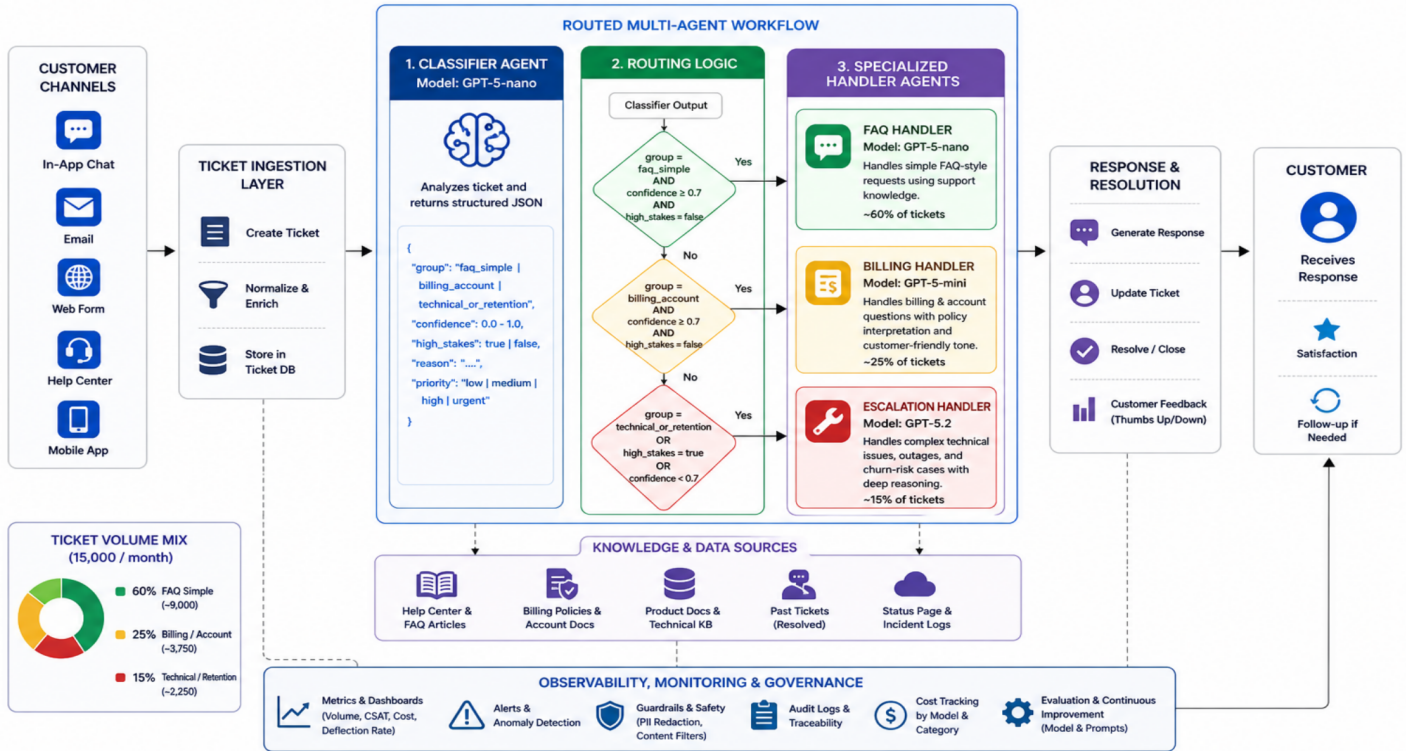
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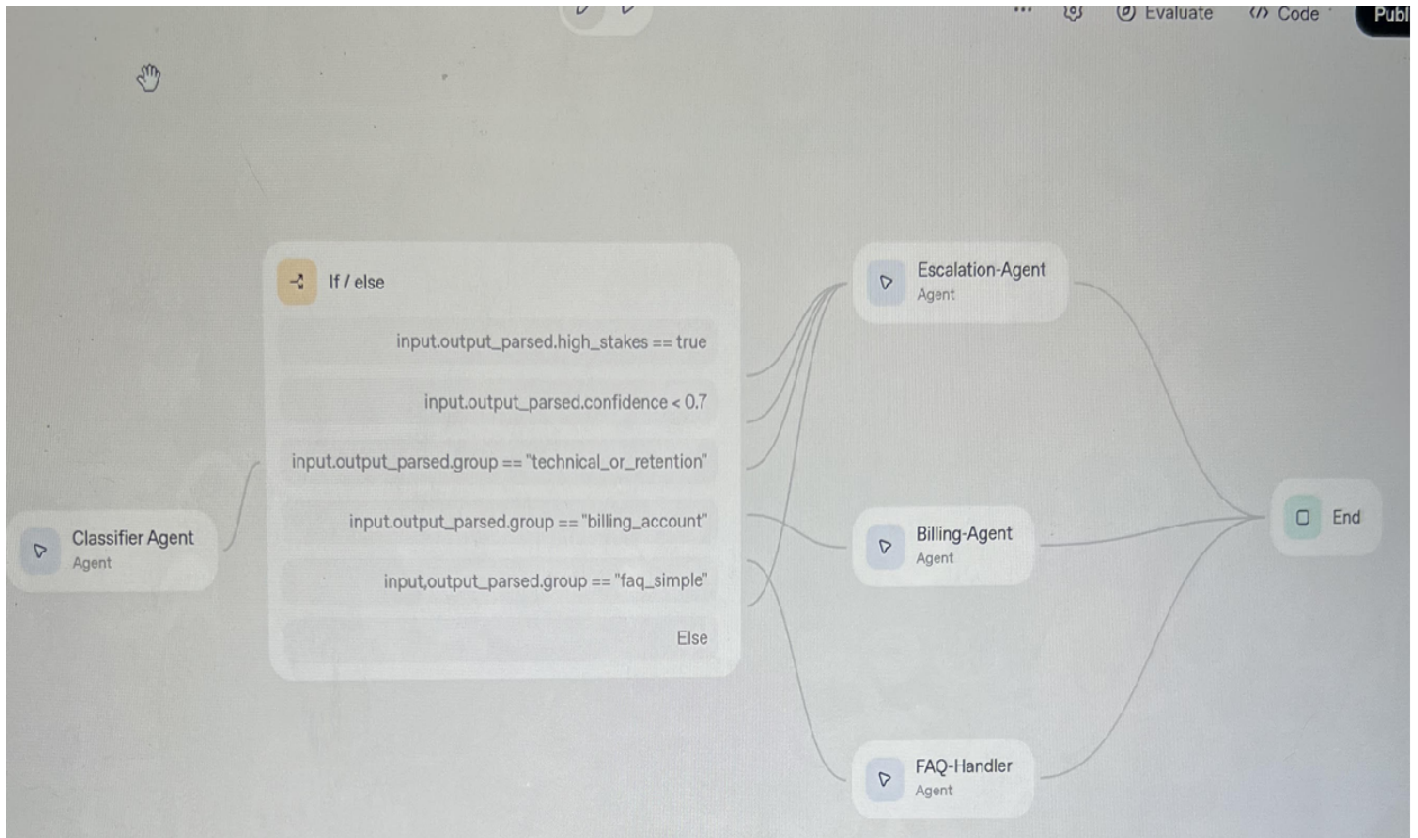
1. Architecture Rationale

CloudSync uses a routed multi-agent architecture to balance cost, performance, and risk. Model Choices: GPT-5-nano handles classification and FAQs (~60%) for speed and low cost. GPT-5-mini handles billing (~25%) with better policy interpretation. GPT-5.2 handles escalation (~15%) for complex and high-risk issues. Routing Logic: Step 1: Classifier determines group, confidence, and risk. Step 2: Conditional routing: - High risk or low confidence → Escalation - Technical issues → Escalation - Billing → Billing agent - FAQ → FAQ agent Why This Works: - Reduces cost by ~70–80% - Improves accuracy by matching model to task - Protects against risk via escalation - Scales efficiently for high ticket volumes Bottom Line: This system intelligently routes requests, optimizes cost, and improves customer experience.

2. Architecture Diagram



3. Working Model



4.1 Test Case 1

CloudSync AI Support Agent

Test Case 1 • Routed to: FAQ-Handler

Classifier route: faq_simple

Responding agent: FAQ-Handler

Customer

Subject: Can't reset my password

Hi, I forgot my password and the reset email isn't coming through. How do I get back into my account?

FAQ-Handler

Response from FAQ-Handler

Hi! Sorry you're locked out—if the reset email isn't coming through, try this:

1. Go to the CloudSync sign-in page
2. Click "Forgot password?"
3. Enter your account email address
4. Look for the reset email, then check your Spam/Junk folder too
5. If you still don't see it, confirm the email address is correct and try requesting the reset again

If it still doesn't arrive after that, I'm going to connect you with a specialist who can help resolve this as quickly as possible.

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4.2 Test Case 2

CloudSync AI Support Agent

Test Case 2 • Routed to: FAQ-Handler

Classifier route: faq_simple

Responding agent: FAQ-Handler

Customer

Subject: Where can I find my invoices?

I need to download our last 3 months of invoices for our finance team. Where do I find them?

FAQ-Handler

Response from FAQ-Handler

Hi! You can download your invoices from your CloudSync billing settings:

1. Sign in to CloudSync
2. Go to Account Settings
3. Select Billing
4. Open Invoice History
5. Download the invoices you need for the last 3 months

That should give your finance team the invoice copies they need.

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4.3 Test Case 3

CloudSync AI Support Agent

Test Case 3 • Routed to: Billing-Agent

Classifier route: billing_account

Responding agent: Billing-Agent

Customer

Subject: Charged twice this month

I just checked my credit card statement, and I see two charges from CloudSync for this month. We are on the Pro plan. Can you fix this?

Billing-Agent

Response from Billing-Agent

Hi, I'm sorry you're seeing two charges this month. For a Pro plan, please check the billing details in your account first:

1. Sign in to CloudSync
2. Go to Account Settings
3. Open Billing
4. Review Invoice History

Compare the invoice entries there with the two charges on your card statement. If they match separate invoices, the charges may be for different billing events; if they appear to be duplicates, the next step is to submit a billing inquiry for review.

I can't directly adjust charges from here, but a billing review will confirm whether one of the charges should be corrected.

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4.4 Test Case 4

CloudSync AI Support Agent

Test Case 4 • Routed to: Escalation-Agent

Classifier route: technical_or_retention

Responding agent: Escalation-Agent

Customer

Subject: Data not syncing for our team

Three people on our team say their data stopped syncing yesterday. I am the workspace owner. We have tried logging out and back in but it is still not working.

Escalation-Agent

Response from Escalation-Agent

Hi, I'm sorry your team is dealing with this sync issue. Since multiple users are affected and logging out and back in did not resolve it, this needs a deeper look.

Please check the following first:

1. Confirm the affected users have a stable internet connection
2. Restart the CloudSync desktop app
3. Make sure the app is updated to the latest version
4. Pause sync, then resume it
5. Verify the correct folders are still selected for syncing
6. Check whether any error messages appear

If the issue continues, please send the number of affected users, the platform they are using, and any error messages so this can be investigated further.

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4.5 Test Case 5

CloudSync AI Support Agent

Test Case 5 • Routed to: Escalation-Agent

Classifier route: technical_or_retention

Responding agent: Escalation-Agent

Customer

Subject: Considering cancellation after repeated issues

This is the third time this quarter we have had problems with our account. Last month we were charged incorrectly, two weeks ago half our team lost access, and now our data is not syncing again. Leadership is reviewing alternatives. We need someone to take ownership of this.

Escalation-Agent

Response from Escalation-Agent

Hi, I understand why leadership is reviewing alternatives. Three serious issues in one quarter—incorrect billing, access loss, and now sync problems again—is not acceptable, and I'm sorry your team has had to deal with that.

I'm going to treat this as a priority and focus on two things right away: stabilizing sync and documenting the full pattern of issues so this gets a deeper review, not another surface-level fix.

Please send me these details so this can be investigated properly:

- Whether the sync issue affects all users or only some
- Whether this is happening on desktop, web, or mobile
- The names of any folders or file types not syncing
- Approximate time the issue started
- Any error messages your team is seeing
- Whether files are delayed, missing, or failing outright

I'll make sure this is handled as a deeper investigation into the repeated failures affecting your account.

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